STRUCTURAL CALCULATION · 3D FRAME · TRAGWERK EXAMPLES

Office building, 3 × 2 bays, 4 storeys

2026-10-22 · Tragwerk 0.3.0 · EN 1992-1-1:2004 with recommended values · 60 nodes, 48 columns, 68 beams ·

MAXIMUM UTILISATION

0.8366

Pass

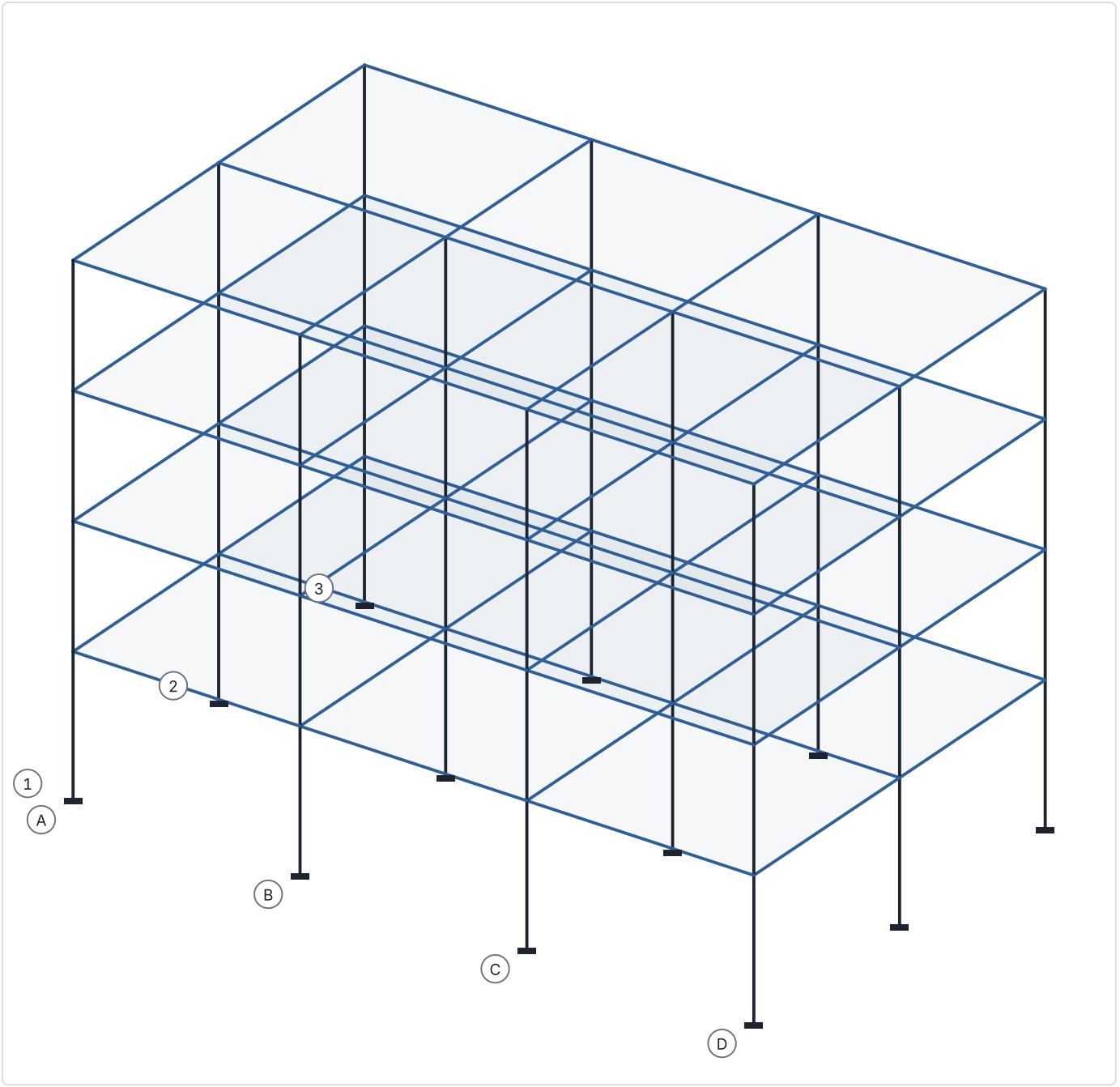
All checks pass

governing: BY-B1-L1, bending with axial force, both axes

1 Basis of design

ITEM	BASIS
Analysis	Linear-elastic 3D frame analysis (direct stiffness method, 6 DOF per node). Members with axial, St Venant torsion and biaxial bending stiffness; gross concrete sections, E = E_{cm}, G = E/(2(1 + ν)), ν = 0.2. Exact member diagrams between nodes.
Slabs	Slabs are not modelled as elements. Their loads reach the beams by tributary areas; no rigid-diaphragm constraint is applied.
Combinations	EN 1990 Eq. (6.10), pattern loading of variable cases, exclusive groups, recommended ψ factors.
Design	EN 1992-1-1:2004 + AC:2010, recommended NDPs: bending with axial force about both axes (6.1, 5.8.9 Eq. 5.39), shear in both planes (6.2), torsion (6.3), slender columns by nominal curvature (5.8.8), detailing (8.2, 9.2, 9.5), deflection (7.4).

2 Structural model



Axonometric view. Dark lines: columns; blue: beams; tinted areas: slab panels (loads only).

GRID	POSITIONS [M]
Axes along X	A = 0, B = 6, C = 12, D = 18
Axes along Y	1 = 0, 2 = 5.5, 3 = 11
Levels Z	L0 = 0, L1 = 3.6, L2 = 6.8, L3 = 10, L4 = 13.2

SECTION	B × H [MM]	CONCRETE [MPA]	LONGITUDINAL BARS	LINKS	MEMBERS
Column	400 × 400	C30/37, f_{cd} = 20.00	top 3Ø20, bottom 3Ø20, each side 1Ø20 (A_s = 2513 mm²)	Ø8 @ 240	48
Beam X	300 × 550	C30/37, f_{cd} = 20.00	top 4Ø20, bottom 3Ø20 (A_s = 2199 mm²)	Ø8 @ 200	36

SECTION	B × H [MM]	CONCRETE [MPA]	LONGITUDINAL BARS	LINKS	MEMBERS
Beam Y	300 × 500	C30/37, f_{cd} = 20.00	top 4Ø20, bottom 3Ø16 (A_s = 1860 mm²)	Ø8 @ 200	32

12 supports, 12 of them fully fixed. Member local axes: x from start to end node; for beams y is horizontal and z points up; for columns z is global X.

3 Actions

CASE	NAME	TYPE	LOADS	ΣF_X [KN]	ΣF_Y [KN]	ΣF_Z [KN]
G1	Self weight of frame and slabs	permanent + self weight	240	0	0	-6 540.6
G2	Finishes and facade	permanent	270	0	0	-2 232.0
Q	Imposed load, all floors	variable, Cat. B, office, ψ = 0.7, 0.5, 0.3	180	0	0	-1 782.0
S	Snow on the roof	variable, Snow, H <= 1000 m, ψ = 0.5, 0.2, 0	60	0	0	-198.0
W+X	Wind in +X	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	12	112.9	0	0
W-X	Wind in -X	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	12	-112.9	0	0
W+Y	Wind in +Y	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	16	0	184.7	0
W-Y	Wind in -Y	variable, Wind, ψ = 0.6, 0.2, 0, group “wind”	16	0	-184.7	0

4 Combinations of actions

74 ULS, 37 characteristic, 12 frequent and 2 quasi-permanent combinations, all checked.

5 Analysis results

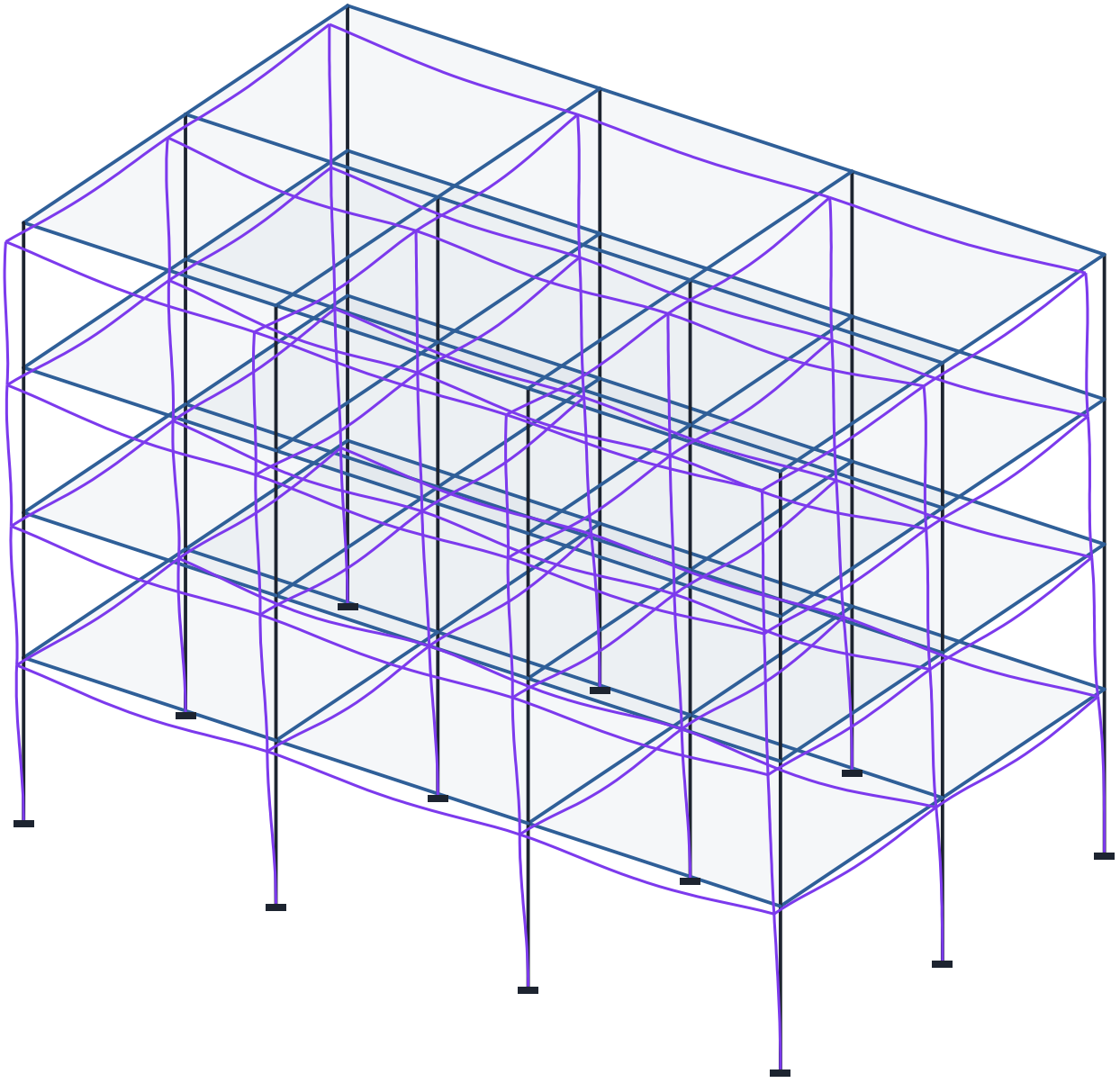
Equilibrium self-check: the largest residual between applied loads and reactions over all load cases is 2.5e-10 kN (or kNm).

Support reactions, ULS envelope (min … max)

NODE	R_X [KN]	R_Y [KN]	R_Z [KN]	M_X [KNM]	M_Y [KNM]	M_Z [KNM]
A1-L0	-3.2 … 25.6	-11.1 … 29.6	428.3 … 754.4	-54.7 … 32.4	-15.2 … 42.4	-0.7 … 0.6
B1-L0	-15.5 … 13.8	-15.3 … 39.8	728.3 … 1291.8	-72.6 … 43.2	-30.0 … 27.9	-0.5 … 0.5
C1-L0	-13.8 … 15.5	-15.3 … 39.8	728.3 … 1291.8	-72.6 … 43.2	-27.9 … 30.0	-0.5 … 0.5
D1-L0	-25.6 … 3.2	-11.1 … 29.6	428.3 … 754.4	-54.7 … 32.4	-42.4 … 15.2	-0.6 … 0.7
A2-L0	-3.7 … 33.7	-23.2 … 23.2	725.6 … 1250.3	-47.0 … 47.0	-18.2 … 54.5	-0.3 … 0.3
B2-L0	-19.0 … 16.8	-30.5 … 30.5	1186.4 … 2086.1	-61.3 … 61.3	-36.6 … 34.0	-0.3 … 0.3

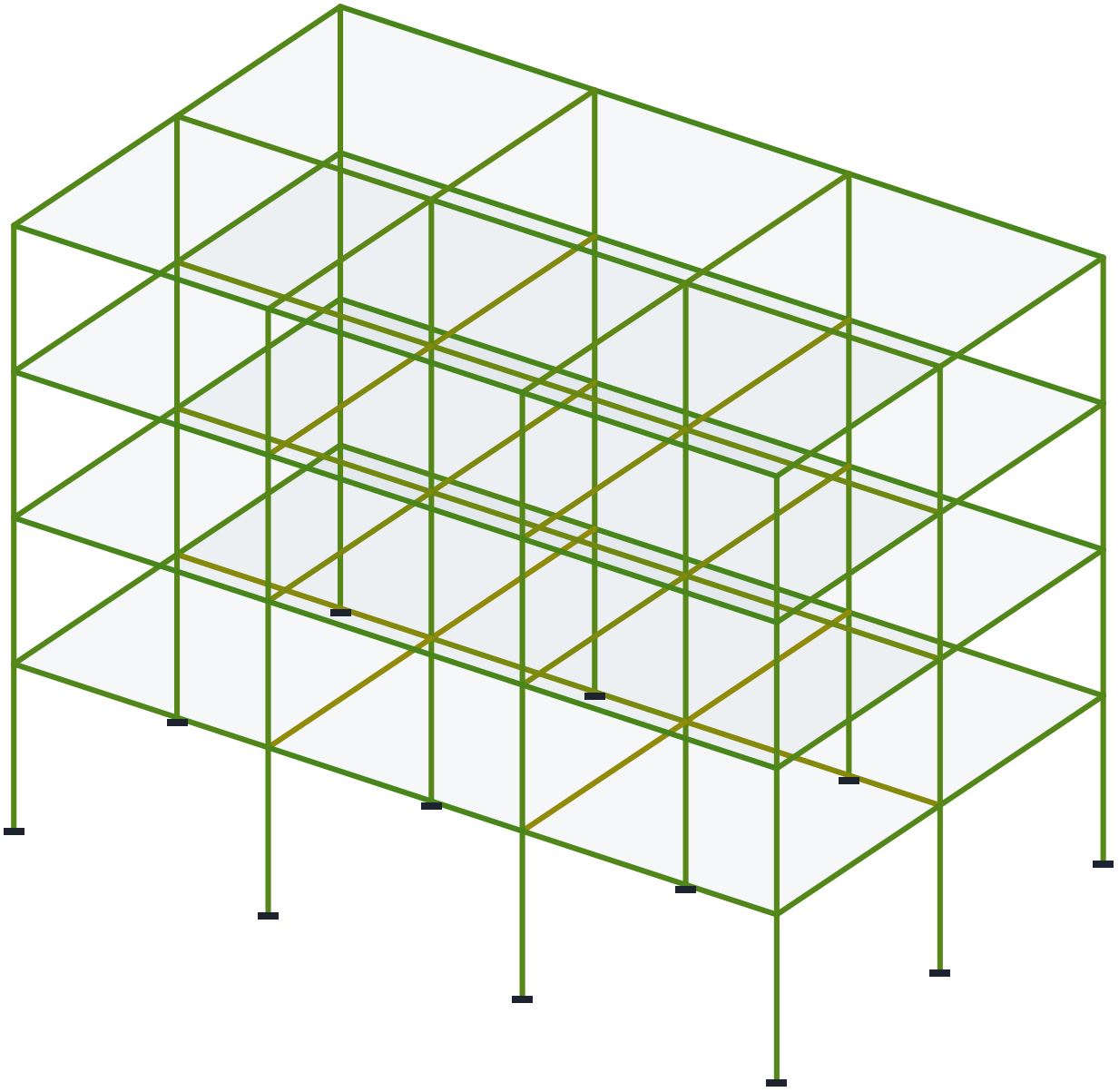
NODE	R_X [KN]	R_Y [KN]	R_Z [KN]	M_X [KNM]	M_Y [KNM]	M_Z [KNM]
C2-L0	-16.8 … 19.0	-30.5 … 30.5	1186.4 … 2086.1	-61.3 … 61.3	-34.0 … 36.6	-0.3 … 0.3
D2-L0	-33.7 … 3.7	-23.2 … 23.2	725.6 … 1250.3	-47.0 … 47.0	-54.5 … 18.2	-0.3 … 0.3
A3-L0	-3.2 … 25.6	-29.6 … 11.1	428.3 … 754.4	-32.4 … 54.7	-15.2 … 42.4	-0.6 … 0.7
B3-L0	-15.5 … 13.8	-39.8 … 15.3	728.3 … 1291.8	-43.2 … 72.6	-30.0 … 27.9	-0.5 … 0.5
C3-L0	-13.8 … 15.5	-39.8 … 15.3	728.3 … 1291.8	-43.2 … 72.6	-27.9 … 30.0	-0.5 … 0.5
D3-L0	-25.6 … 3.2	-29.6 … 11.1	428.3 … 754.4	-32.4 … 54.7	-42.4 … 15.2	-0.7 … 0.6

Deflected shape, CHAR035
max. displacement 5.0 mm, drawn ×190



Short-term elastic deflections, gross sections.

Largest utilisation per member
green: low, amber: close to 1, red: fails



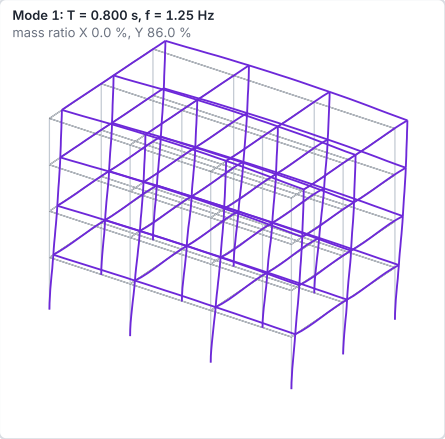
6 Dynamic analysis

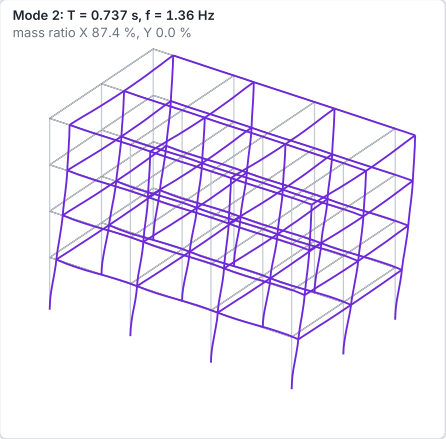
Eigenvalue problem $(K - \omega^2 M)\varphi = 0$ on a model with every member divided into 2 elements; consistent element mass matrices $M_e = \int \bar{m}\, N^T N \, dx$; modes normalised to $\varphi^T M \varphi = 1$. Total mass 948.7 t.

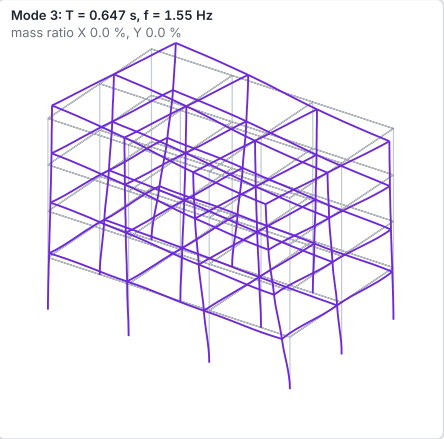
LOAD CASE	MASS FACTOR
G1	1.000
G2	1.000
Q	0.3000
S	0
W+X	0
W-X	0
W+Y	0

LOAD CASE	MASS FACTOR
W-Y	0

MODE	T [S]	F [HZ]	$M_{EFF,X}$ [%]	$M_{EFF,Y}$ [%]	$M_{EFF,Z}$ [%]	Σ X [%]	Σ Y [%]
1	0.7997	1.250	0.0	86.0	0.0	0.0	86.0
2	0.7374	1.356	87.4	0.0	0.0	87.4	86.0
3	0.6471	1.545	0.0	0.0	0.0	87.4	86.0
4	0.4326	2.312	0.0	0.0	0.0	87.4	86.0
5	0.3711	2.695	0.0	0.8	0.0	87.4	86.8
6	0.2785	3.590	0.5	0.0	0.0	88.0	86.8
7	0.2716	3.682	0.0	8.4	0.0	88.0	95.2
8	0.2642	3.785	0.0	0.0	0.0	88.0	95.2
9	0.2519	3.970	7.9	0.0	0.0	95.9	95.2
10	0.2347	4.261	0.0	0.0	0.0	95.9	95.2
11	0.1974	5.066	0.0	0.0	0.0	95.9	95.2
12	0.1888	5.297	0.0	0.8	0.0	95.9	96.0







7 Summary of design checks

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BY-B1-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BY-C1-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BY-B2-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BY-C2-L1	beam	Beam Y	Bending with axial force, both axes	0.8366	Pass
BX-A2-L1	beam	Beam X	Bending with axial force, both axes	0.7945	Pass
BX-C2-L1	beam	Beam X	Bending with axial force, both axes	0.7945	Pass
BY-C1-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass
BY-C2-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass
BY-B1-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BY-B2-L3	beam	Beam Y	Bending with axial force, both axes	0.7822	Pass
BY-B1-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BY-C2-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BY-C1-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BY-B2-L2	beam	Beam Y	Bending with axial force, both axes	0.7683	Pass
BX-B2-L1	beam	Beam X	Bending with axial force, both axes	0.7508	Pass
BX-A2-L2	beam	Beam X	Bending with axial force, both axes	0.7204	Pass
BX-C2-L2	beam	Beam X	Bending with axial force, both axes	0.7204	Pass
BX-B2-L2	beam	Beam X	Bending with axial force, both axes	0.7116	Pass
BX-A2-L3	beam	Beam X	Bending with axial force, both axes	0.7038	Pass
BX-C2-L3	beam	Beam X	Bending with axial force, both axes	0.7038	Pass
BX-B2-L3	beam	Beam X	Bending with axial force, both axes	0.6926	Pass
BY-C1-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
BY-B1-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
BY-B2-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
BY-C2-L4	beam	Beam Y	Bending with axial force, both axes	0.6349	Pass
C-A1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L1	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
C-C2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L2	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L3	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D1-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D2-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-A3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-B3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-C3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
C-D3-L4	column	Column	Maximum link spacing in columns	0.6000	Pass
BY-A1-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L1	beam	Beam Y	Maximum link spacing	0.5874	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BY-A1-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L2	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A1-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L3	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A1-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-A2-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D1-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BY-D2-L4	beam	Beam Y	Maximum link spacing	0.5874	Pass
BX-A2-L4	beam	Beam X	Bending with axial force, both axes	0.5828	Pass
BX-C2-L4	beam	Beam X	Bending with axial force, both axes	0.5828	Pass
BX-A3-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-C3-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-C1-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-A1-L1	beam	Beam X	Bending with axial force, both axes	0.5685	Pass
BX-B2-L4	beam	Beam X	Bending with axial force, both axes	0.5403	Pass
BX-B1-L1	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L1	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A1-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B1-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C1-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A3-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C3-L2	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A1-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B1-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C1-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A3-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C3-L3	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A1-L4	beam	Beam X	Maximum link spacing	0.5312	Pass

MEMBER	TYPE	SECTION	GOVERNING CHECK	UTILISATION H	RESULT
BX-B1-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C1-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-A3-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-B3-L4	beam	Beam X	Maximum link spacing	0.5312	Pass
BX-C3-L4	beam	Beam X	Maximum link spacing	0.5312	Pass

8 Design checks in detail

Shown for every member that fails and the 12 most utilised members. Run with the full option for every member; all members appear in the summary above.

Member BY-B1-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 Pass

Governing at x = 2.475 m from node i; combination ULS028: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.124	**-0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 0.000 m from node i; combination ULS019: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W-X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C1-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 — Pass

Governing at x = 2.475 m from node i; combination ULS028: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.124	**0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 0.000 m from node i; combination ULS011: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W+X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the top (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B2-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 — Pass

Governing at x = 3.025 m from node i; combination ULS036: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.124	**-0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 0.000 m from node i; combination ULS019: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W-X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the top (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C2-L1 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.8366 — Pass

Governing at x = 3.025 m from node i; combination ULS036: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-Y

N_{Ed}	axial force (compression +) = -11.67	**-11.67** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 92.45	**92.45** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.124	**0.1235** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**110.7** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**92.69** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.835^{1.00} + 0.00133^{1.00}]^{1/1.00}$	**0.8366**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6132 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 136.3	**136.3** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -11593.8/150\,000$	**-0.0773** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0773] \cdot 300.0 \cdot 452.0$	**80.51** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.3/222.3$	**0.6132**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0044

Pass

Governing at x = 0.000 m from node i; combination ULS011: 1.35·G1 + 1.35·G2 + 1.05·Q + 0.75·S + 1.50·W+X

V_{Ed}	design shear force = 0.494	**0.4937** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -19553.5/150\,000$	**-0.1304** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.13] \cdot 500.0 \cdot 226.7$	**71.64** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.494/111.5$	**0.0044**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1324

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5513**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.913** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.913/22.00	**0.1324**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BX-A2-L1 · Beam X · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7945 — Pass

Governing at x = 6.000 m from node i; combination ULS010: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+X

N_{Ed}	axial force (compression +) = -15.03	**-15.03** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = -199.5	**-199.5** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -5.242e-15	**-5.242e-15** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**-251.1** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**108.2** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 165 000·20.00 + 2199.1·434.8	**4 256.1** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.795^{1.00} + 0^{1.00}]^{1/1.00}$	**0.7945**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6323 Pass

Governing at x = 6.000 m from node i; combination ULS010: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+X

V_{Ed}	design shear force = 156.1	**156.1** kN	
d	effective depth to the top (tension) reinforcement	**502.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/502.0)}$	**1.631**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 502.0)$	**0.0083**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -15030.0/165\,000$	**-0.0911** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.631^{1.5} \cdot 30.00^{0.5}$	**0.3994** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.631 \cdot (100 \cdot 0.00834 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0911] \cdot 300.0 \cdot 502.0$	**84.18** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 502.0$	**451.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 451.8 \cdot 434.8 \cdot 2.500$	**246.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 451.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**493.6** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**246.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 156.1/246.8$	**0.6323**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0076

Pass

Governing at x = 0.000 m from node i; combination ULS022: 1.35·G1 + 1.35·G2 + 1.50·W+Y

V_{Ed}	design shear force = 0.862	**0.8624** kN	
d	effective depth to the bottom (tension) reinforcement	**229.3** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/229.3)}$	**1.934**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 942.5/(550.0 \cdot 229.3)$	**0.0075**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -18211.0/165\,000$	**-0.1104** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.934^{1.5} \cdot 30.00^{0.5}$	**0.5155** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.934 \cdot (100 \cdot 0.00747 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.11] \cdot 550.0 \cdot 229.3$	**80.44** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·229.3	**206.4** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 206.4 \cdot 434.8 \cdot 2.500$	**112.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 550.0 \cdot 206.4 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**413.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**112.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.862/112.8$	**0.0076**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3332

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 165\,000$	**6 600.0** mm²
$A_{s,prov}$	all longitudinal bars	**2 199.1** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1805 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5312 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 502.0$	**376.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.5054

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0063**	
ρ'	compression reinforcement ratio	**0**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/(\rho - \rho') + (1/12)\cdot\sqrt{f_{ck}}\cdot\sqrt{(\rho'/\rho_0)}]$, ρ' taken as 0 (conservative; the equation is singular as ρ' → ρ)	**23.65**	Eq. 7.16b
l/d	span/effective depth = 6000.0/502.0	**11.95**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1394

Pass

Governing at x = 3.000 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**6 109.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**2 694.5** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**62.96** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5702**	Eq. 7.19
w_{max}	$\int \kappa\cdot\bar{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**3.345** mm	Eq. 7.18
w_{lim}	span/250 = 6000.0/250	**24.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 3.345/24.00	**0.1394**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BX-C2-L1 · Beam X · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7945 — Pass

Governing at x = 0.000 m from node i; combination ULS018: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-X

N_{Ed}	axial force (compression +) = -15.03	**-15.03** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = -199.5	**-199.5** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -6.361e-15	**-6.361e-15** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**-251.1** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**108.2** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 165 000·20.00 + 2199.1·434.8	**4 256.1** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.795^{1.00} + 0^{1.00}]^{1/1.00}$	**0.7945**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.6323 Pass

Governing at x = 0.000 m from node i; combination ULS018: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-X

V_{Ed}	design shear force = 156.1	**156.1** kN	
d	effective depth to the top (tension) reinforcement	**502.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/502.0)}$	**1.631**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ = 1256.6/(300.0·502.0)	**0.0083**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = -15030.0/165 000	**-0.0911** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.631^{1.5} \cdot 30.00^{0.5}$	**0.3994** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.631 \cdot (100 \cdot 0.00834 \cdot 30.00)^{1/3} +$ 0.15·-0.0911]·300.0·502.0	**84.18** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·502.0	**451.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot \theta$ = 0.503·451.8·434.8·2.500	**246.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd} / (\cot \theta + \tan \theta)$ = 1.0·300.0·451.8·0.528·20.00/(2.500 + 0.4)	**493.6** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**246.8** kN	6.2.1
η	V_{Ed}/V_{Rd} = 156.1/246.8	**0.6323**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0076

Pass

Governing at x = 0.000 m from node i; combination ULS022: 1.35·G1 + 1.35·G2 + 1.50·W+Y

V_{Ed}	design shear force = 0.862	**0.8624** kN	
d	effective depth to the top (tension) reinforcement	**229.3** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/229.3)}$	**1.934**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 942.5/(550.0 \cdot 229.3)$	**0.0075**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -18211.0/165\,000$	**-0.1104** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.934^{1.5} \cdot 30.00^{0.5}$	**0.5155** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.934 \cdot (100 \cdot 0.00747 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.11] \cdot 550.0 \cdot 229.3$	**80.44** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 229.3$	**206.4** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 206.4 \cdot 434.8 \cdot 2.500$	**112.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 550.0 \cdot 206.4 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**413.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**112.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.862/112.8$	**0.0076**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3332

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 165\,000$	**6 600.0** mm²
$A_{s,prov}$	all longitudinal bars	**2 199.1** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1805 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5312 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 502.0$	**376.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ \text{mm})$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ \text{mm})$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.5054

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0063**	
ρ'	compression reinforcement ratio	**0**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/(\rho - \rho') + (1/12)\cdot\sqrt{f_{ck}}\cdot\sqrt{(\rho'/\rho_0)}]$, ρ' taken as 0 (conservative; the equation is singular as ρ' → ρ)	**23.65**	Eq. 7.16b
l/d	span/effective depth = 6000.0/502.0	**11.95**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1394

Pass

Governing at x = 3.000 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**6 109.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**2 694.5** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**62.96** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5702**	Eq. 7.19
w_{max}	$\int \kappa\cdot\bar{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**3.345** mm	Eq. 7.18
w_{lim}	span/250 = 6000.0/250	**24.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 3.345/24.00	**0.1394**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C1-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS020: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.087	**-0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566

Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 123.7/222.3$	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 0.000 m from node i; combination ULS031: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W-Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the top (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C2-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS020: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.087	**-0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 123.7/222.3$	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 0.000 m from node i; combination ULS023: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W+Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B1-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS012: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.087	**0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·452.0	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} = 123.7/222.3	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077

Pass

Governing at x = 0.000 m from node i; combination ULS031: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W-Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the bottom (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B2-L3 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7822 — Pass

Governing at x = 2.750 m from node i; combination ULS012: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+X

N_{Ed}	axial force (compression +) = -7.633	**-7.633** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.11	**87.11** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.087	**0.0870** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**111.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**93.10** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 0	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.781^{1.00} + 9.350e\text{-}04^{1.00}]^{1/1.00}$	**0.7822**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5566 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 123.7	**123.7** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 5119.6/150 000	**0.0341** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0341] \cdot 300.0 \cdot 452.0$	**82.78** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 123.7/222.3$	**0.5566**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0077 — Pass

Governing at x = 0.000 m from node i; combination ULS023: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W+Y

V_{Ed}	design shear force = 0.856	**0.8564** kN	
d	effective depth to the top (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ $= -5384.4/150\,000$	**-0.0359** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot -0.0359] \cdot 500.0 \cdot 226.7$	**73.25** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd} / (\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00 / (2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.856/111.5$	**0.0077**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100 — Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 150\,000$	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1102

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, $\beta = 0.5$, at the governing station	**0.5315**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.424** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.424/22.00	**0.1102**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-B1-L2 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7683 — Pass

Governing at x = 2.750 m from node i; combination ULS012: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W+X

N_{Ed}	axial force (compression +) = 2.267	**2.267** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.13	**87.13** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = 0.0789	**0.0789** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**113.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**94.09** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 5.952e-04	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.767^{1.00} + 8.388e\text{-}04^{1.00}]^{1/1.00}$	**0.7683**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5799 Pass

Governing at x = 5.500 m from node i; combination ULS026: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W+Y

V_{Ed}	design shear force = 128.9	**128.9** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 13833.8/150 000	**0.0922** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0922] \cdot 300.0 \cdot 452.0$	**83.96** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·452.0	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} = 128.9/222.3	**0.5799**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0067

Pass

Governing at x = 0.000 m from node i; combination ULS007: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W+X

V_{Ed}	design shear force = 0.751	**0.7505** kN	
d	effective depth to the top (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 2509.0/150 000	**0.0167** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0167] \cdot 500.0 \cdot 226.7$	**74.15** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 0.751/111.5$	**0.0067**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ = 0.04·150 000	**6 000.0** mm²	
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²	

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1275

Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5364**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\, dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.804** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.804/22.00	**0.1275**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member BY-C2-L2 · Beam Y · beam

Bending with axial force, both axes · EN 1992-1-1 6.1, 5.8.9 · η = 0.7683 — Pass

Governing at x = 2.750 m from node i; combination ULS020: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.75·S + 0.90·W-X

N_{Ed}	axial force (compression +) = 2.267	**2.267** kN	
M_{Edy}	moment in the x-z plane (compression on +z face +) = 87.13	**87.13** kNm	
M_{Edz}	moment in the x-y plane (compression on +y face +) = -0.0789	**-0.0789** kNm	
M_{Rdy}	uniaxial resistance at N_{Ed}, x-z plane (strain compatibility, Fig. 6.1)	**113.5** kNm	6.1
M_{Rdz}	uniaxial resistance at N_{Ed}, x-y plane	**-94.09** kNm	6.1
N_{Rd}	$A_c \cdot f_{cd} + A_s \cdot f_{yd}$ = 150 000·20.00 + 1859.8·434.8	**3 808.6** kN	5.8.9(4)
a	1.0 / 1.5 / 2.0 at N_{Ed}/N_{Rd} = 0.1 / 0.7 / 1.0, interpolated = N_{Ed}/N_{Rd} = 5.952e-04	**1.000**	5.8.9(4)
η	$[(M_{Edy}/M_{Rdy})^a + (M_{Edz}/M_{Rdz})^a]^{1/a}$ = $[0.767^{1.00} + 8.388e\text{-}04^{1.00}]^{1/1.00}$	**0.7683**	Eq. 5.39

- Eq. 5.39 is applied to every member; with bending about one axis only it reduces to the uniaxial check of 6.1.

Shear, x-z plane (V_{z}) · EN 1992-1-1 6.2 · η = 0.5799 Pass

Governing at x = 0.000 m from node i; combination ULS034: 1.35·G1 + 1.35·G2 + 1.50·Q + 0.90·W-Y

V_{Ed}	design shear force = 128.9	**128.9** kN	
d	effective depth to the top (tension) reinforcement	**452.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/452.0)}$	**1.665**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 1256.6/(300.0 \cdot 452.0)$	**0.0093**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 13833.8/150 000	**0.0922** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.665^{1.5} \cdot 30.00^{0.5}$	**0.4119** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.665 \cdot (100 \cdot 0.00927 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0922] \cdot 300.0 \cdot 452.0$	**83.96** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 452.0$	**406.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 406.8 \cdot 434.8 \cdot 2.500$	**222.3** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 406.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**444.4** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**222.3** kN	6.2.1
η	V_{Ed}/V_{Rd} = 128.9/222.3	**0.5799**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Shear, x-y plane (V_{y}) · EN 1992-1-1 6.2 · η = 0.0067

Pass

Governing at x = 0.000 m from node i; combination ULS015: 1.35·G1 + 1.35·G2 + 0.75·S + 1.50·W-X

V_{Ed}	design shear force = 0.751	**0.7505** kN	
d	effective depth to the top (tension) reinforcement	**226.7** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/226.7)}$	**1.939**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 829.4/(500.0 \cdot 226.7)$	**0.0073**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 2509.0/150 000	**0.0167** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.939^{1.5} \cdot 30.00^{0.5}$	**0.5177** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.939 \cdot (100 \cdot 0.00732 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0.0167] \cdot 500.0 \cdot 226.7$	**74.15** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	$0.9 \cdot d$ $= 0.9 \cdot 226.7$	**204.1** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 204.1 \cdot 434.8 \cdot 2.500$	**111.5** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 500.0 \cdot 204.1 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**371.5** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**111.5** kN	6.2.1
η	V_{Ed}/V_{Rd} = 0.751/111.5	**0.0067**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.3100

Pass

$A_{s,max}$	$0.04 \cdot A_c$ = 0.04·150 000	**6 000.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 859.8** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.3401 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 454.0$	**205.1** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**603.2** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.1625 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 452.0$	**204.2** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**1 256.6** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5874 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 454.0$	**340.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2386 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 16.00)/2$	**88.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(16.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.4375 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 4 \cdot 20.00)/3$	**48.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.4020 — Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0044**	
ρ'	compression reinforcement ratio	**0.0092**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/\rho + 3.2\cdot\sqrt{f_{ck}}\cdot(\rho_0/\rho - 1)^{3/2}]$	**30.13**	Eq. 7.16a
l/d	span/effective depth = 5500.0/454.0	**12.11**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.1275 — Pass

Governing at x = 2.750 m from node i; combination QP002: 1.00·G1 + 1.00·G2 + 0.30·Q

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**4 416.2** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**1 514.8** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**48.75** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0.5364**	Eq. 7.19
w_{max}	$\int \kappa\cdot\tilde{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.804** mm	Eq. 7.18
w_{lim}	span/250 = 5500.0/250	**22.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.804/22.00	**0.1275**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

9 Assumptions and limitations

- Linear-elastic first-order analysis with gross sections; slender columns get second-order moments by the nominal curvature method on the isolated member (5.8.8) with the effective lengths set in the model.
- The generator assumes a braced frame (effective length = storey height). For an unbraced sway frame, increase the column effective length factors.
- Slabs are not designed and not modelled as plates; there is no rigid-diaphragm constraint. Wind is applied at the windward facade nodes.
- Biaxial bending by EN 1992-1-1 Eq. 5.39 using exact uniaxial resistances; shear checked in each plane separately; torsion shown where $T_{Ed} \geq 0.1\cdot T_{Rd,c}$.
- Reinforcement uniform along each member; anchorage, laps, crack widths and punching are not checked.
- Seismic design (where present) is a DCL response spectrum analysis: no capacity design, no accidental torsion, no vertical component; P-Δ sensitivity θ is reported, not applied.
- A licensed engineer remains responsible for the model, its inputs and the use of these results.

10 Software verification

> This build ran its verification suite when producing this report: 49 benchmarks with 232 comparisons, **all passed**.

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